THE POOP APP

Team Members: Alejandro Fonseca, Carlos Fernández-Yáñez, Pablo Simón Martín and Álvaro Martín

Group: 196

Teacher: Boni García Gutiérrez

1. Introduction

The Poop App is a funny yet useful mobile application developed for Android devices using Kotlin in Android Studio. Its core functionality allows users to log their bathroom visits and track their digestive habits over time, helping them become more aware of their health in a lighthearted way. However, it doesn't stop at simple tracking—The Poop App introduces a social and gamified layer that makes the experience more engaging. Users can create or join leagues with friends and see if they manage to get on top of the leaderboard.

In addition, the app includes a map feature that helps users locate and review nearby public bathrooms, making it especially handy while traveling. The application leverages several Firebase services, such as Firebase Authentication for secure login or Cloud Firestore for storing and syncing poop logs and user data. With real-time updates, users can instantly see changes in leaderboards or log activity. Ultimately, The Poop App delivers a unique blend of health monitoring, social interaction, and toilet humor, encouraging users to engage with their well-being in an enjoyable and relatable way.

2. Distribution of Work

Throughout the project, we maintained close collaboration to ensure efficient development. Work was evenly distributed, and each member took leadership of a core feature of the application, this is, each member led the design and implementation of a key feature:

- Alejandro was responsible for most of the Google Maps integration, enabling users to view and interact with nearby bathroom locations directly on the map.

- Carlos focused on Firebase integration and user authentication, ensuring that users could sign up, log in, and securely store data in the cloud.

- Pablo developed the Poop Tracker and Notifications system, which allowed users to log their bathroom visits and receive reminders or alerts.

- Álvaro led the creation of Shitshare, the app's social feature.

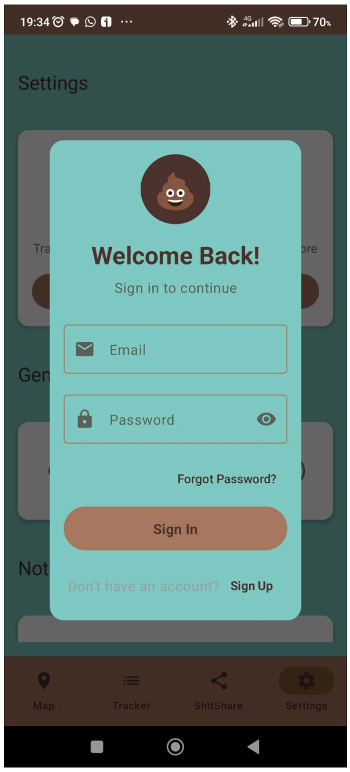
To coordinate their work, the team used GitHub for version control. We followed a branch-based workflow, where each feature was developed in a separate branch and later merged into the main project through pull requests. This organized and collaborative approach ensured a balanced workload and resulted in a well-structured and fully functional application.

3. Main Features

The Poop App includes several core features that define its functionality and user experience. Each feature was designed with a focus on simplicity and enjoyment, while leveraging Firebase and Android technologies in the background. Below are the main features:

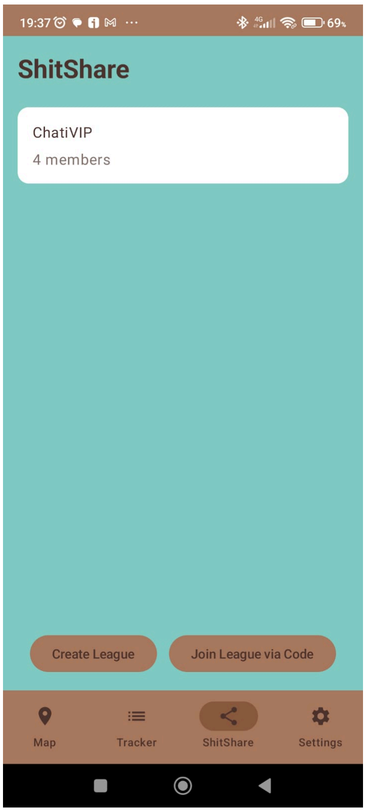
3.1 Authentication (Login & Register)

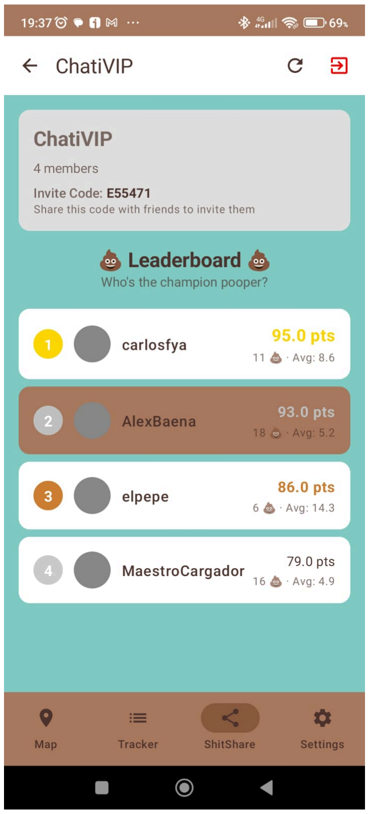
User authentication is provided via a Firebase Authentication integration. New users can register an account using email and password, and returning users can securely log in. Firebase Authentication handles account creation and sign-in, including identity management and encoding. This feature includes form validation for inputs (like valid email format and password strength) and feedback for login errors. Upon successful login or sign-up, the user is authenticated and can access the app's main functionalities.



3.2 Leaderboard (League Rankings)

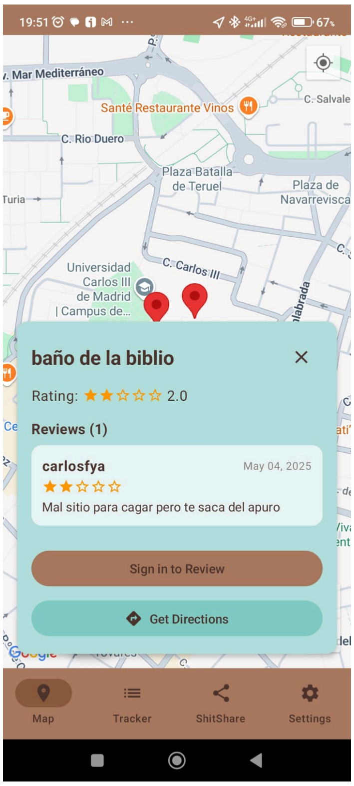
A gamified **leaderboard** displays user rankings within a league (a group of friends or invited users). The leaderboard ranks members based on the number of logged "poops" or a scoring system derived from their activity. This creates a friendly competition, as users strive to climb the ranks by logging more events. The data for the leaderboard is stored in Firebase and updates in real-time – whenever a member logs a new entry, all users in the league see the updated rankings immediately. This real-time update fosters a sense of competition and engagement among friends. Users can either create their own league, or join a legue via code.

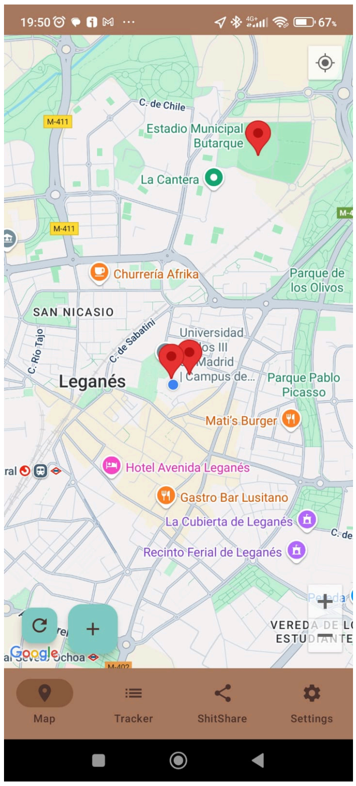




3.3 Map Screen (Find Nearby Bathrooms)

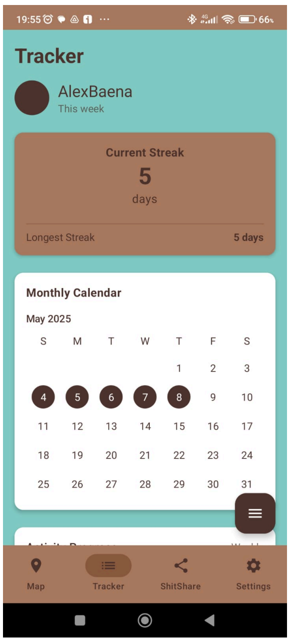
The app includes a Map screen that helps users find and review nearby bathrooms using the Google Maps API. It shows the user's location and public restroom markers. Users can tap markers for details, add new locations, rate restrooms, or get directions via Google Maps. This feature promotes a community-driven approach to identifying clean or convenient bathrooms. It enhances the app's practical value and expands its purpose beyond personal tracking to shared contributions, making the experience more useful and engaging for everyone.

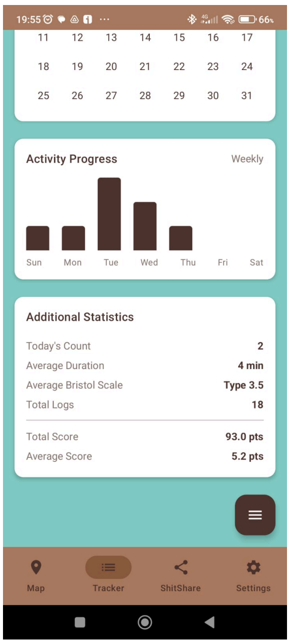


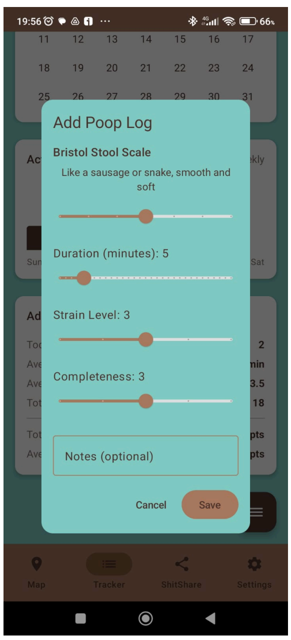


3.4 Poop Tracker (Logging Events)

At the heart of the application is the **Poop Tracker**, which allows users to log each bathroom visit as an event. When recording a "poop," the user can input details such as strain level and completeness, the **duration** of the bathroom visit, and any **notes** or comments. Entries from the Poop Tracker feed into the statistics for habit tracking – over time, users can observe patterns (for example, frequency per day) and into the leaderboard scoring.

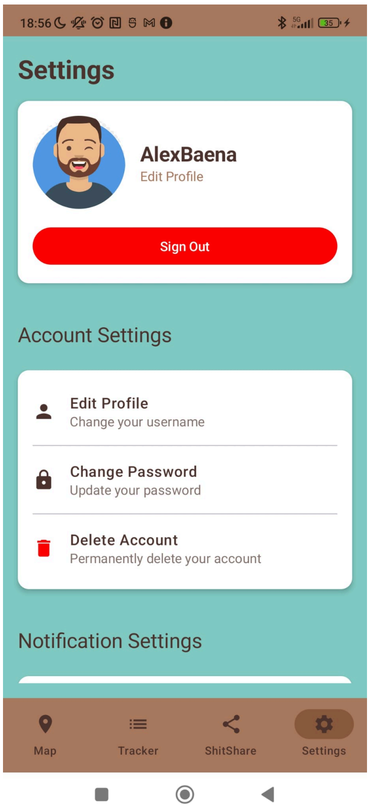


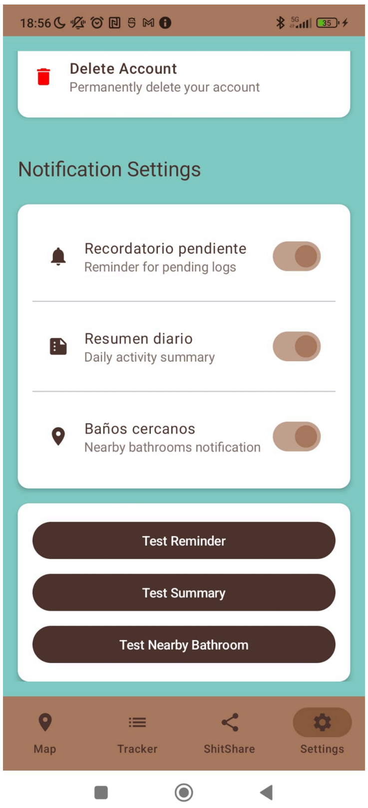


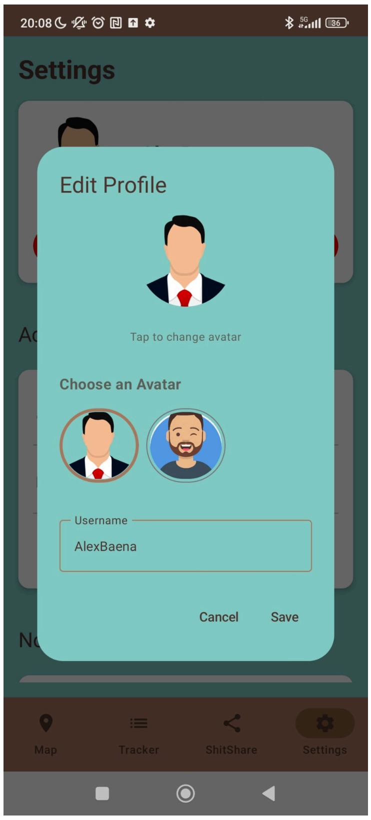


3.5 Settings Screen

Each user has a Profile section where they can view and edit personal information. The profile screen displays the user's username and avatar. Users can edit their display username and change their avatar. Changing the profile picture will display a set of avatars that the user can select. The profile section also provides options such as logging out. In this part, we can also find a Test-Mode of the app, where you can find buttons to test the notifications. Test reminder automatically sends a notification that reminds you to go to the bathroom. The other two notification buttons have not been completely implemented, as we did not have time to complete those functionalities. However, we left the buttons there as we plan to finish the app this summer for fun.







4. Secondary Features (Technical Implementation)

In addition to the primary user-facing features, The Poop App incorporates several extra functionalities. Below is an overview of the third party libraries, automated tests and other secondary features of the project.

- **Firebase Firestore**: The app uses Cloud Firestore as its cloud database for storing structured data. Firestore is a flexible NoSQL database that keeps data in sync across clients through real time listeners. In our implementation, Firestore holds users' log entries, profile info, and league data (such as members and their scores). This allows the leaderboard and other shared data to update instantly for all users. Firestore's offline support ensures the app remains functional even without network connectivity, queuing data updates for sync when back online.

- **Jetpack Compose**: The entire user interface is built with Jetpack Compose. It allows us to construct the UI using declarative Kotlin code instead of XML layouts. Using Compose, we implemented dynamic layouts that react to state changes (for instance, updating the UI when new data comes in).

- **DataStore (Preferences)**: For local data persistence, the app uses Jetpack DataStore to save simple preferences. DataStore is a Jetpack library that replaces SharedPreferences for storing key-value data or typed objects. In The Poop App, DataStore holds user preferences such as app settings or maybe a toggle (for example, whether the user wants to receive notifications or has seen an onboarding tutorial). Using DataStore improves reliability and consistency of saved preferences over the legacy SharedPreferences approach.

- **Notification Manager**: The app includes functionality for local notifications to remind or engage users (for instance, a reminder to go to the bathroom). For this, we use Android's NotificationManager system service to schedule and display notifications. The NotificationManager is the Android service that allows apps to show notifications in the status bar. In our implementation, we create a notification channel and use the NotificationManager to issue notifications at appropriate times (e.g., a daily reminder).

- **Coroutines**: The app's architecture leverages Kotlin Coroutines and StateFlow extensively for managing asynchronous tasks and UI state. Coroutines are used to perform operations like network calls (Firebase interactions) or file I/O off the main thread, preventing the UI from freezing – they provide a concise way to write asynchronous code that is both efficient and easy to read.

5. Project Source Code ZIP File

https://drive.google.com/drive/folders/1_rdT8H8-gu4czgesDdgBESArS9tjtzW7?usp=drive_link

As in the mid review we did not send the zip of the project correctly, we have also shared another zip, just in case zipping from Android Studio was not working correctly or we made an error again. This zip was done by just manually exporting the files to a zip in the file manager.

https://drive.google.com/drive/folders/11ieS2JDBb-Svo6xSYIwRW_XtsTAwwaS0?usp=drive_link

6. AVD Considerations

For optimal testing, we recommend running the app on a Pixel 4 emulator with Android 11 (API Level 30 with Google Play), which was our primary development environment. This setup ensures full compatibility with Jetpack Compose and Firebase libraries, and provides smooth graphics and animation performance. While the app is compatible with other devices, Pixel 4 @ API 30 offers the most stable results.

Additionally, make sure the SDK includes: Android 7.0 (Nougat), Google APIs, Android SDK Platform 24, and both Intel x86 and x86_64 Atom system images.

7. Demo Video

Demonstration of The Poop App:

https://youtu.be/hUigo7fBZ8A

The following videos we made them with an AI just for fun, we included them because we find them funny:

- Official promotional advertisement of the app: https://www.youtube.com/shorts/VES_N5N8YoI?feature=share
- Official song of the app: https://youtu.be/flzQ3S6S8_w

8. Summary and Outlook

In summary, The Poop App project successfully met the objectives that were set at the beginning of the course. The team built a fully functional Android application that demonstrates the control of development practices and libraries seen in class. Through features like real-time leaderboards, mapping, and robust logging, we delivered an experience that is both entertaining and useful. The integration of Firebase services and Jetpack Compose ensures the app is scalable and maintainable for future growth.

9. References

1. Android Developers – *Jetpack Compose* (Official documentation)developer.android.com.
2. Android Developers – *Jetpack DataStore* (Official documentation on DataStore preferences)developer.android.com.
3. Firebase Documentation – *Firebase Authentication* (Product overview and integration)firebase.google.com.
4. Firebase Documentation – *Cloud Firestore* (NoSQL database overview)firebase.google.com.
5. Android Developers – *Kotlin Coroutines on Android* (Guide for using coroutines in Android apps)
6. AI Assistants for coding- *Cursor and Chagpt*